Standard Errors in Empirical Data

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How often are estimators defined?

This is across 50 outcomes. Each column shows the percent of time the listed estimator is not defined when using the given covariate set. To calculate, I first averaged across the outcomes within a set, and then equally weighted these outcomes across sets. For most datasets, either all the outcomes or none of the outcomes were defined for a given model.

Estimator	na_ATE_best	na_ATE_all	na_SE_none	na_SE_best	na_SE_all
LRa-FIcw-het	0.08	0.27	0.62	0.62	0.75
LRa-FIcw-db	0.08	0.27	0.62	0.62	0.73
LRapw-FIpw-db	0.08	0.27	0.62	0.62	0.73
LRapw-FIpw-het	0.08	0.27	0.62	0.62	0.73
LRi-FIpw-db	0.08	0.27	0.62	0.62	0.73
LRicw-FIcw-db	0.08	0.27	0.62	0.62	0.73
LRi-FIpw-crve	0.00	0.19	0.00	0.00	0.19
LRicw-FIcw-crve	0.00	0.19	0.00	0.00	0.19
LRa-FE-het	0.00	0.00	0.00	0.00	0.13
LRapw-FE-het	0.00	0.00	0.00	0.00	0.05
LRa-FE-db	0.00	0.00	0.00	0.00	0.04
LRapw-FE-db	0.00	0.00	0.00	0.00	0.04
LRi-FE-db	0.00	0.00	0.00	0.00	0.04
LRicw-FE-db	0.00	0.00	0.00	0.00	0.04
LRi-FE-crve	0.00	0.00	0.00	0.00	0.00
LRicw-FE-crve	0.00	0.00	0.00	0.00	0.00
MLM-FE	0.00	0.00	0.00	0.00	0.00
MLM-FIcw	0.00	0.00	0.00	0.00	0.00
MLM-RE	0.00	0.00	0.00	0.00	0.00

What predicts larger or smaller estimated standard errors?

We run a regression of the log of the estimated standard error on the estimator, the type of estimator, degrees of freedom, sample size, and the covariate option used in the estimation.

We restrict our attention to only those outcomes where all estimators have defined estimated standard errors for a given level of covariate adjustment.

This subsetting takes us from 50 outcomes to 16 outcomes. It also removes some extreme outliers of the estimated standard errors.

Our regression model is of the form:

```
Msimp <- lmer( log_SE ~ Covariates +
                 df + J + nbar + I(n/100) + I(100*cluster_ICC) + I(100*block_ICC) +
                 (1|UniqueID) + (1|UniqueID:Covariates) + (1|Estimator),
               data = cc )
```

```
## Warning: Some predictor variables are on very different scales: consider
## rescaling
```

```
M <- update( Msimp, . ~ . +  Estimand + type + FI + db + crve )
```

```
## Warning: Some predictor variables are on very different scales: consider
## rescaling
```

We fit one model with just random effects for the estimators, and another with aspects of the estimators to see if certain qualities are associated with higher or lower estimated standard errors.

And the results are, when fit to all our estimates:

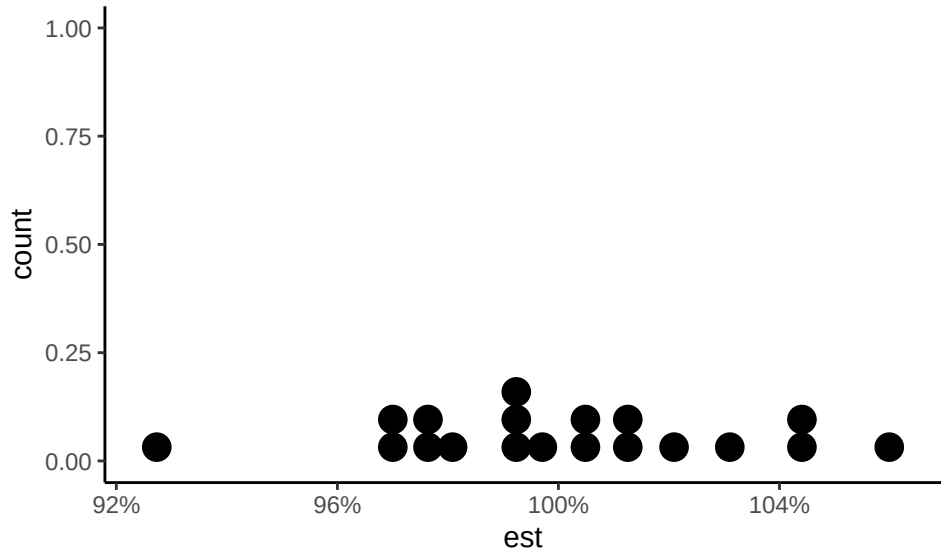
term	est	CIl	CIh	log_est	std.error	statistic
(Intercept)	0.08	0.05	0.12	-2.51	0.20	-12.67
Covariatesbest	0.79	0.72	0.87	-0.23	0.05	-4.79
Covariatesall	0.80	0.71	0.90	-0.23	0.06	-3.70
df	1.00	1.00	1.00	0.00	0.00	-0.69
J	1.00	1.00	1.00	0.00	0.00	0.38
nbar	1.01	1.00	1.02	0.01	0.00	1.95
I(n/100)	0.99	0.97	1.00	-0.01	0.01	-2.03
I(100 * cluster_ICC)	1.02	1.00	1.04	0.02	0.01	1.72
I(100 * block_ICC)	1.01	0.97	1.04	0.01	0.02	0.30
EstimandPerson	0.95	0.94	0.97	-0.05	0.01	-5.79
typeLRi	1.07	1.04	1.09	0.06	0.01	5.63
typeMLM	0.92	0.90	0.95	-0.08	0.01	-6.26
FI	0.99	0.97	1.00	-0.01	0.01	-1.93
db	0.94	0.92	0.96	-0.06	0.01	-5.19
crve	0.91	0.88	0.94	-0.10	0.02	-6.04

I exponentiated all the coefficients so they are multiplicative factors. A factor of 1.1 means a 10% increase over baseline, and a 0.9 means a 10% decrease.

Generally covariates reduce standard errors by 20% on average. Some of the estimators are systematically smaller (the crve estimators, for example).

The model explains 97% of the variance in the log of the estimated standard errors.

We can also see if some estimators are systematically more varying than others:



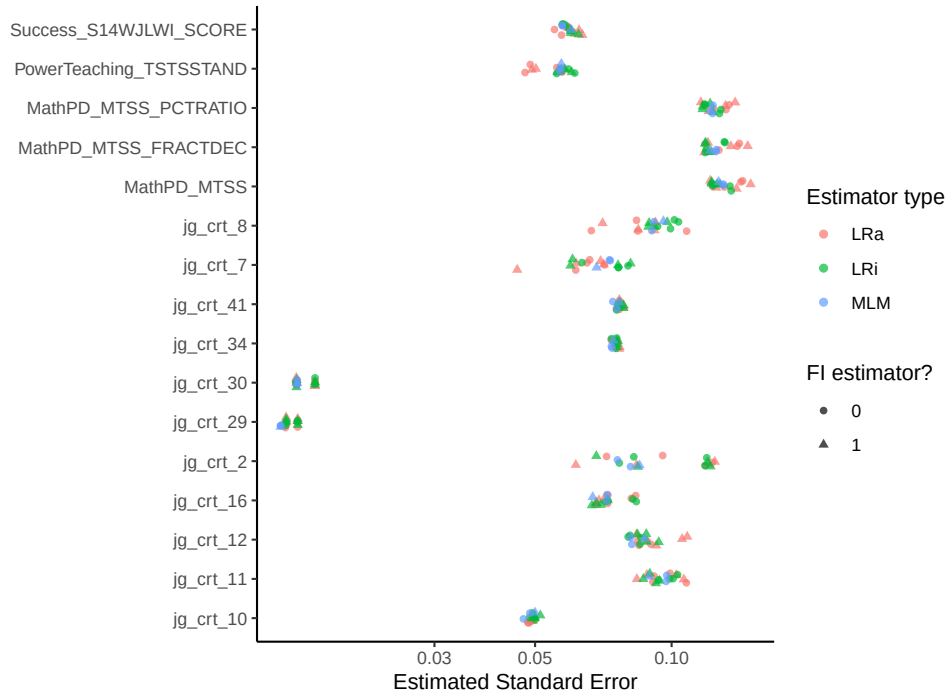
Estimator	est	sd_resid
LRicw-FIcw-db	1.06	1.56
LRicw-FE-db	1.04	0.85
LRa-FE-het	1.04	0.97
LRa-FIcw-het	1.03	0.98
LRapw-FE-het	1.02	0.84
LRi-FE-db	1.01	0.92
LRa-FE-db	1.01	0.69
LRicw-FIcw-crve	1.01	0.82
LRicw-FE-crve	1.00	0.69
LRapw-FIpw-het	1.00	0.91
LRi-FIpw-db	0.99	1.53
LRa-FIcw-db	0.99	1.59
LRi-FE-crve	0.99	0.72
MLM-FIcw	0.98	0.49
MLM-FE	0.98	0.49
LRi-FIpw-crve	0.97	0.70
MLM-RE	0.97	0.62
LRapw-FE-db	0.97	0.98
LRapw-FIpw-db	0.93	1.55

Overall, the estimated standard errors across estimators are, on average, about the same. The highest is about 15% larger than the lowest, averaged across all the outcomes.

We can also look at whether the estimators have different residuals, which would give a sense of stability. That is the third column on the table above. In this column I calculated the ratio of the standard deviation of the residuals (in log space) divided by the overall standard deviation of the residuals. This is both a measure of the instability of the estimator and also a measure of how well the model fits the data for that estimator, so it is unclear how reliable these numbers are. That said, some estimators have much wider ranges of residuals than others.

Looking at all the standard errors

We can plot all the standard errors to look at grouping. We do this for the 'best' covariate set.



We then replicate the regression from above, but just for the "best" covariate set.

A regression table of the four regressions considered is as follows, with the point estimates exponentiated for interpretation:

term	core	all	best	simple
(Intercept)	0.08	0.08	0.07	0.07
Covariatesbest	0.79	0.79		
Covariatesall	0.80	0.80		
df	1.00	1.00	1.00	1.00
J	1.00	1.00	1.00	1.00
nbar	1.01	1.01	1.01	1.01
I(n/100)	0.99	0.99	0.99	0.99
I(100 * cluster_ICC)	1.02	1.02	1.02	1.02
I(100 * block_ICC)	1.01	1.01	1.00	1.00
EstimandPerson		0.95	0.94	0.96
typeLRi		1.07	1.05	
typeMLM		0.92	0.91	
FI		0.99	0.99	1.00
db		0.94	0.92	0.99
crve		0.91	0.91	